

Classroom Electricity Management API Documentation

Version: 1.0

Communication: HTTP

Base URL

http://<ESP_IP>

Replace <ESP_IP> with the actual IP address assigned to the system on your network.

1. Control Electricity Supply

Endpoint:

GET /relay

Query Parameters

Parameter	Type	Required	Description
apikey	String	 Yes	Your private API key (Do not share this publicly).
room	String	 Yes	The room identifier to control (e.g., VIB109, VIB110).
state	String	 Yes	The desired state: ON or OFF.

Example Request

GET http://<ESP_IP>/relay?apikey=YOUR_API_KEY&room=VIB109&state=ON

Response Codes & Messages

Status Code	Message
200 OK	"Electricity supply turned ON for room VIB109"
200 OK	"Electricity supply turned OFF for room VIB109"
400 Bad Request	"Missing 'room' or 'state' parameter!"
400 Bad Request	"Invalid room number! Check the available rooms."
400 Bad Request	"Invalid state! Use ON or OFF only."
403 Forbidden	"Unauthorized! Invalid API key."

2. Error Handling

- If an **invalid API key** is used, the server will respond with 403 Forbidden.
- If a **missing or incorrect parameter** is provided, the server will return 400 Bad Request.
- If an **unsupported room number** is specified, the server will return "Invalid room number!".

3. Usage Notes

- The system should be **connected to the network** and have a **static or known IP address**.
- Ensure proper API authentication by keeping the API key secure.
- Use **proper HTTP encoding** for special characters in URLs.